1 **(a)** Gradient $= -2$ **[B1]**

y-intercept $= 4$ **[B1]**

$-2x + 4$ **[B1]**

(b) $2x + y = 4$ oe **[B1]**

[Total Marks: 4]

2 **(a)** $\dfrac{(3a+2)-(a+4)}{(a-1)-2a} = -4$ **[M1]**

$\dfrac{2a-2}{-a-1} = -4$ **[M1]**

$2a - 2 = 4a + 4$

-6 = 2a

a = -3[$A1$]

(b) Identifies coordinates of A as $(-6, 1)$ or B as $(-4, -7)$ **[M1]**

$y - 1 = -4(x + 6)$ or $y = -4x - 23$ **[M1]**

$4x + y = -23$ **[A1]**

(c) $4x + (3x - 2) = -23$ or $-4x - 23 = 3x - 2$ oe **[M1]**

$7x = -21$

x = -3[$A1$]

$y = -11$ **[A1]**

[Total Marks: 9]

3 **(a)** Rearranges equation to $y = mx + c$ form: $5y = -3x + 15$ or $y = -\dfrac{3}{5}x + 3$ **[M1]**

Gradient $= -\dfrac{3}{5}$ **[A1]**

y-intercept $= 3$ **[A1]**

(b) Identifies parallel gradient as $-\dfrac{3}{5}$ **[M1]**

$-\dfrac{3}{5}x - 4$ **[A1]**

[Total Marks: 5]

4 **(a)** Calculates the midpoint of $AB = \left(\dfrac{-3+5}{2}, \dfrac{1+5}{2}\right) = (1, 3)$ **[B1]**

Calculates the gradient of $AB = \dfrac{5-1}{5-(-3)} = \dfrac{4}{8} = \dfrac{1}{2}$ **[B1]**

Finds the perpendicular gradient $= -2$ **[M1]**

Uses perpendicular gradient -2 and midpoint $(1, 3)$ to find equation, e.g., $3 = -2(1) + c$ **[M1]**

$-2x + 5$ **[A1]**

(b) Sets up simultaneous equations, e.g., substituting $y = -2x + 5$ into $x - 2y = 5$ **[M1]**

$x - 2(-2x + 5) = 5 \implies 5x - 10 = 5 \implies x = 3$ **[M1]**

$(3, -1)$ **[A1]**

(c) Uses distance formula for $A(-3, 1)$ and $P(3, -1)$ **[M1]**

$\sqrt{(3 - (-3))^2 + (-1 - 1)^2}$ **[M1]**

$\sqrt{40}$ or $2\sqrt{10}$ **[A1]**

[Total Marks: 10]

5 **(a)** Gradient $= \dfrac{-2 - 3}{6 - (-4)}$ or $\dfrac{-5}{10} = -\dfrac{1}{2}$ **[M1]**

Substitution of $(-4, 3)$ or $(6, -2)$ into $y = -\dfrac{1}{2}x + c$, e.g. $3 = -\dfrac{1}{2}(-4) + c$ **[M1]**

$-\dfrac{1}{2}x + 1$ **[A1]**

(b) $C = (5, 0)$ **[B1]**

$D = (0, -3)$ **[B1]**

(c) $x = -4$ **[B1]**

[Total Marks: 6]

6 **(a)** Identifies two correct points on the line, e.g., $(0, 4)$ and $(2, 7)$ **[M1]**

Calculates the gradient $m = \dfrac{7 - 4}{2 - 0} = \dfrac{3}{2}$ **[M1]**

States correct equation in required integer form, e.g., $3x - 2y = -8$ or $2y - 3x = 8$ **[A1]**

(b) Uses $m_1 \times m_2 = -1$ to find gradient of M: $-\dfrac{2}{3}$ **[M1]**

Substitutes $(9, -1)$ into $y = -\dfrac{2}{3}x + c$ to find c: $-1 = -\dfrac{2}{3}(9) + c$ **[M1]**

$-\dfrac{2}{3}x + 5$ **[A1]**

[Total Marks: 6]